Ex: 3	Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer
Date:	

Aim

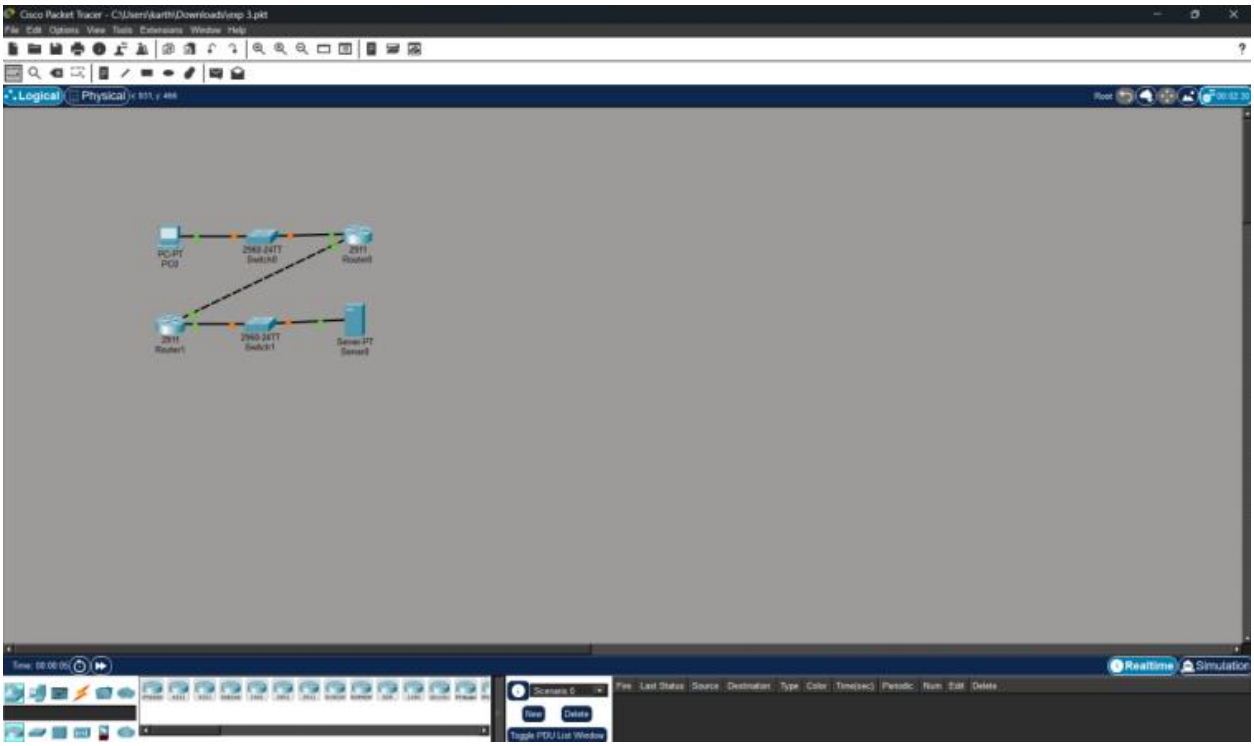
The objective of this simulation is to establish a secure hybrid cloud connection by linking an on-premises data center with public cloud infrastructure. This is achieved using routers, switches, and static routing protocols within the Cisco Packet Tracer environment.

What is Hybrid Cloud?

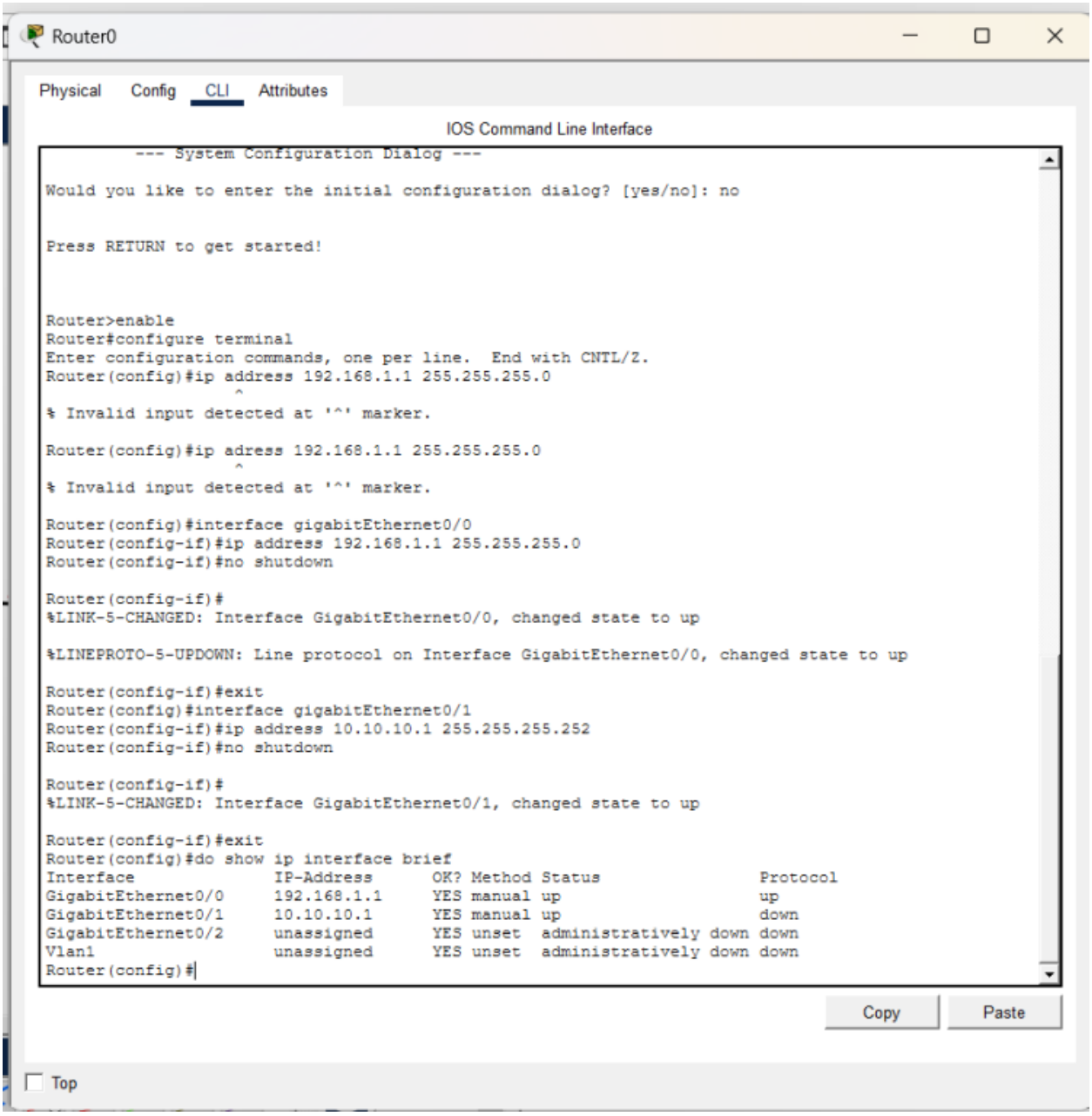
A hybrid cloud architecture combines private, on-premises infrastructure with public cloud services. This integration allows for seamless data and application portability, enabling organizations to leverage the benefits of both environments effectively.

Procedure

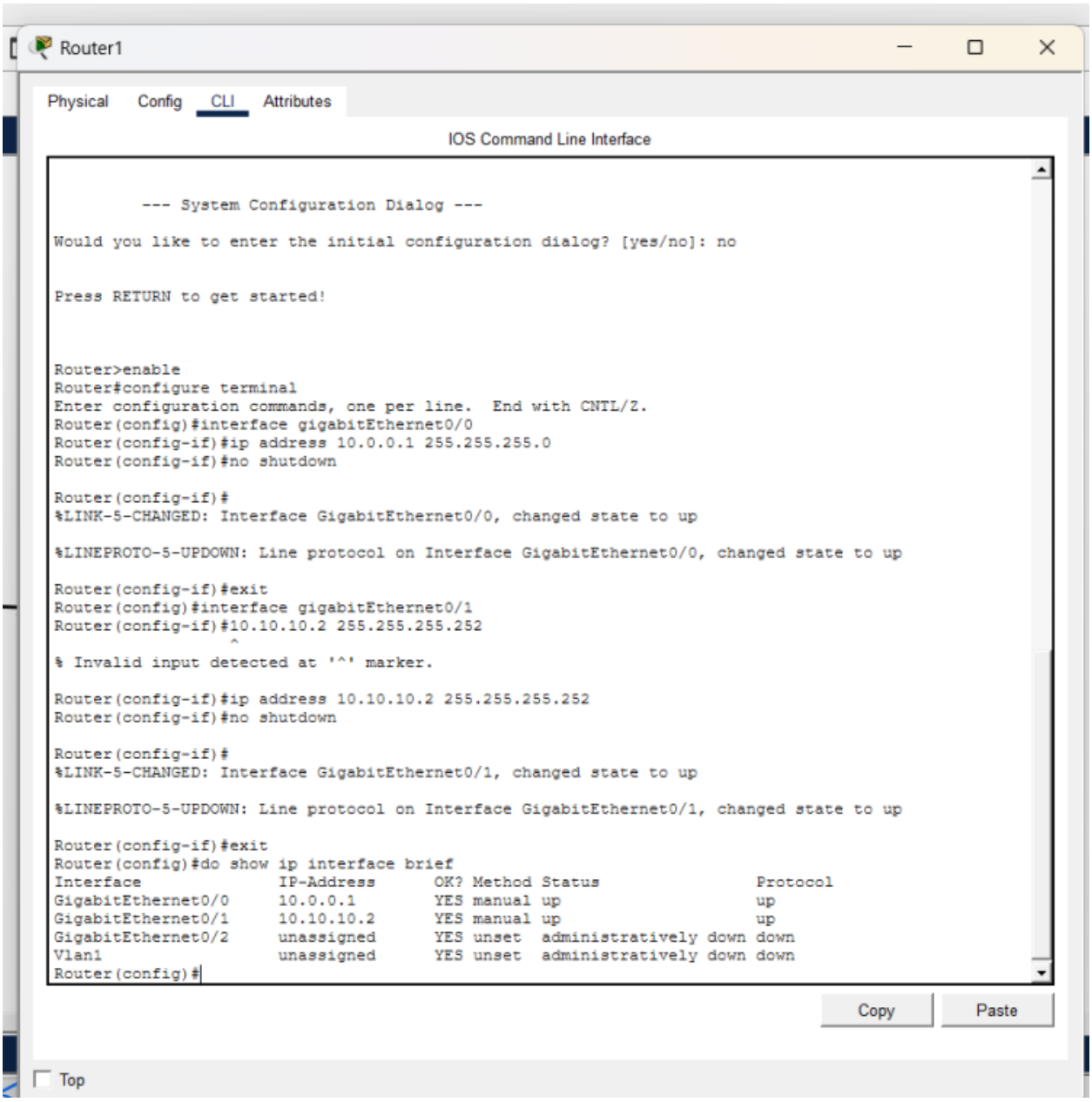
1. Design the Network Topology: Create a layout in Cisco Packet Tracer using two routers, two switches, one PC, and one server, all connected via Copper Straight-Through cables.



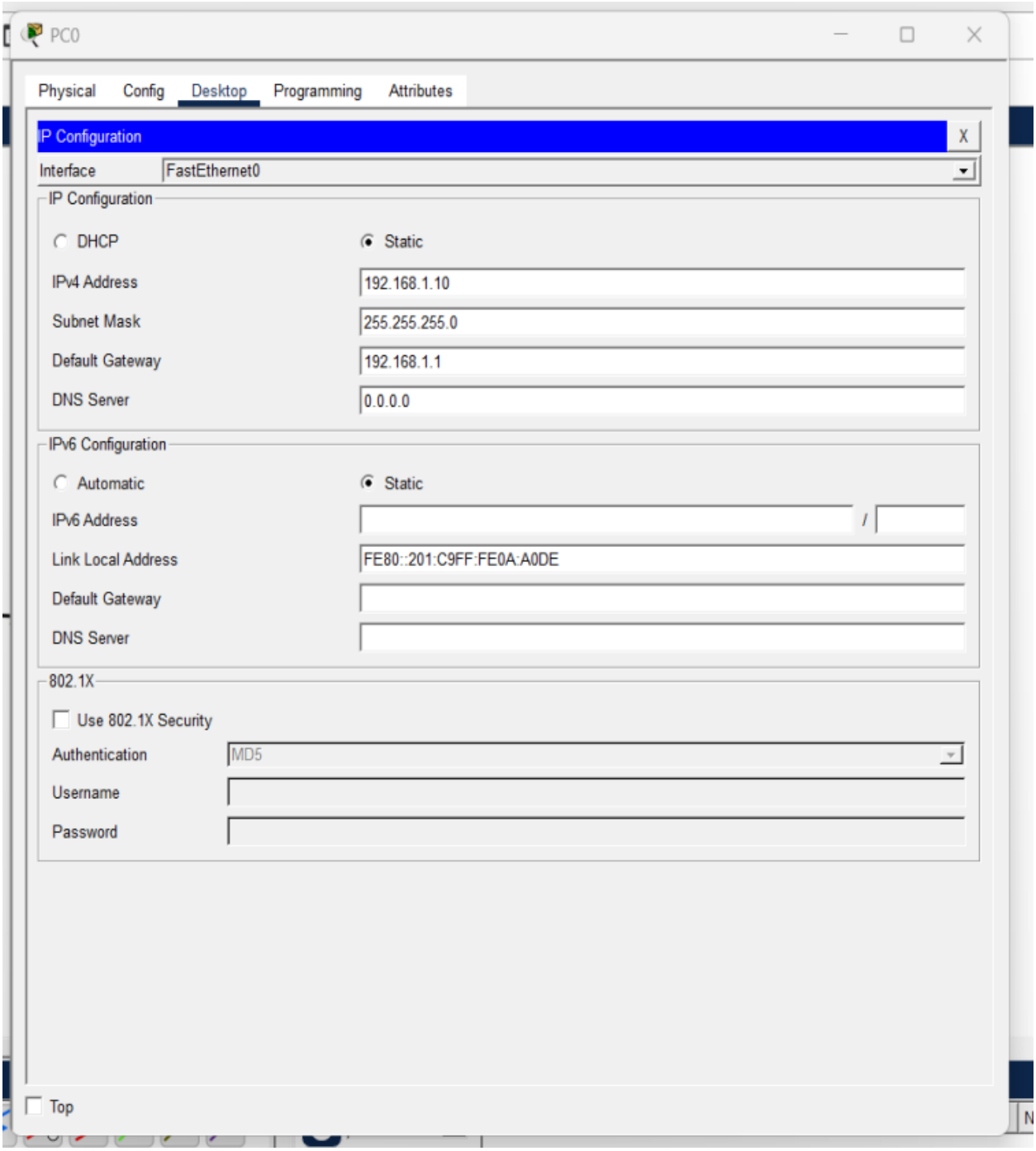
2. Configure Router0:
    a. Set Interface GigabitEthernet0/0 to IP 192.168.1.1 (Subnet Mask: 255.255.255.0).
    b. Set Interface GigabitEthernet0/1 to IP 10.10.10.1 (Subnet Mask: 255.255.255.252).
    c. Activate both interfaces using the 'no shutdown' command.



3. Configure Router1:
    a. Set Interface GigabitEthernet0/0 to IP 10.0.0.1 (Subnet Mask: 255.255.255.0).
    b. Set Interface GigabitEthernet0/1 to IP 10.10.10.2 (Subnet Mask: 255.255.255.252).
    c. Activate both interfaces using the 'no shutdown' command.



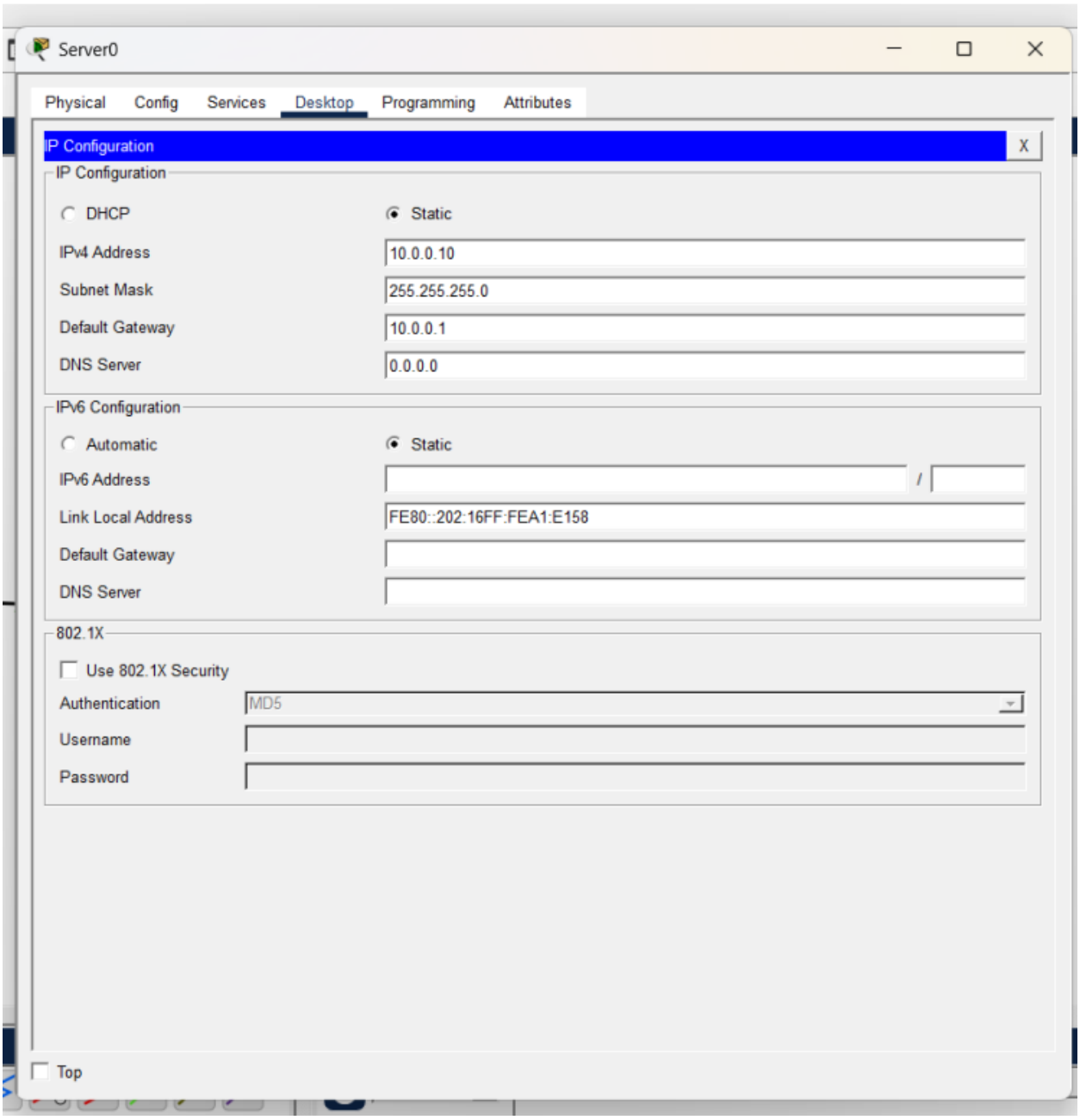
4. Verify Interfaces: Check the status of all configured interfaces with the 'do show ip interface brief' command.
5. Configure PC0 (On-Premises): Assign IP 192.168.1.10, Subnet Mask 255.255.255.0, and set the Default Gateway to 192.168.1.1.

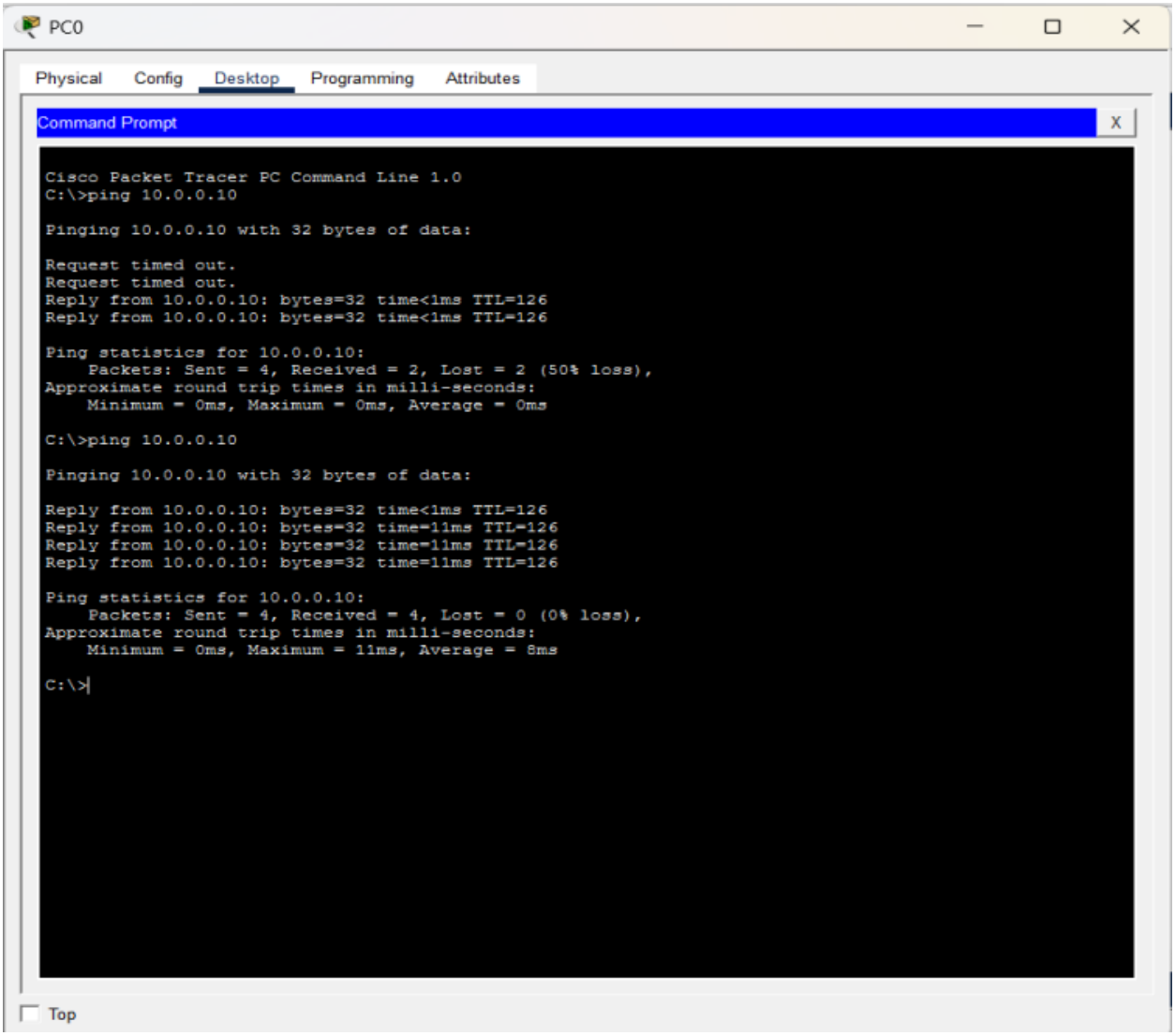


6. Configure Server0 (Cloud): Assign IP 10.0.0.10, Subnet Mask 255.255.255.0, and set the Default Gateway to 10.0.0.1.

7. Implement Static Routing:
    a. On Router0, configure a route to the 10.0.0.0/24 network via 10.10.10.2.
    b. On Router1, configure a route to the 192.168.1.0/24 network via 10.10.10.1.
8. Save Configurations: Ensure settings persist across reboots using the 'copy running-config startup-config' command.

9. Validate the Routing Table: Execute 'show ip route' to confirm that the static routes have been correctly added.
10. Test End-to-End Connectivity: Open the command prompt on PC0 and ping the Cloud Server at 10.0.0.10 to verify communication.





11. Finalize: Save the simulation project as a .pkt file.

Conclusion

The simulation successfully demonstrated hybrid cloud connectivity. By utilizing static routing in Cisco Packet Tracer, the local network and cloud environment were effectively integrated, allowing for reliable resource communication.